Software Design Description 004

Quantization Noise Characterization

Brief

The compression pipeline (SDD501) produces visible noise in reconstructed frames. This document lays out the DSP theory behind quantization noise so the noise sources can be identified, measured, and tuned.

The goal is not to eliminate noise – it's to understand where it comes from, quantify it, and set pipeline parameters (q_step, top_k, seq_thresh, block_threshold) based on measured data instead of guesses.

Background – Quantization Noise (Lyons Ch. 12)

When you round a continuous value to a discrete level, the rounding error is called **quantization noise**. For a uniform quantizer with step size q:

- The error e is uniformly distributed in $[-q/2, +q/2]$
- The noise **variance** (power) is:

$$\sigma_q^2 = \frac{q^2}{12}$$

- The **SNR** of a b-bit quantizer at full scale is:

$$\text{SNR}_{\max} = 6.02 \cdot b + 1.76 \text{ dB}$$

The key intuition: **every bit of precision buys you ~6 dB of SNR**. Doubling the step size (losing 1 bit) costs 6 dB. Halving it gains 6 dB.

Noise Sources in the Pipeline

The pipeline has three stages that destroy information. Each one introduces its own noise. They stack – the total reconstruction error is the sum of all three.

1. Coefficient Quantization (`q_step`)

```
quantized = round(coeff / q_step)
reconstructed = quantized * q_step
```

This is a textbook uniform quantizer. With `q_step = 32`:

Quantity	Value
Step size q	32
Max rounding error	+/-16
Noise variance per coeff	$32^2/12 = 85.3$
Equivalent bits	$\log_2(32) = 5$ bits lost

After IWHT, the noise from each quantized coefficient spreads across all 64 pixels in the 8x8 block. The IWHT divides by 64, so per-pixel noise variance from a single coefficient's quantization error is:

$$\sigma_{\text{pixel}}^2 = \frac{q^2}{12 \cdot 64} = \frac{32^2}{768} \approx 1.33$$

That's per coefficient. With `top_k` coefficients contributing, the total per-pixel quantization noise variance is roughly:

$$\sigma_{\text{total}}^2 \approx k \cdot \frac{q^2}{768}$$

For k=5, q=32: $\sigma^2 \approx 6.7$, so $\sigma \approx 2.6$ gray levels RMS.

That seems small, but it's **per changed block per frame**. Over time, these errors accumulate in the reference frame because the decoder adds the reconstructed diff to the previous frame.

2. Top-K Truncation (`top_k`)

After sequency masking with `seq_thresh = 2`, there are up to 6 nonzero coefficient positions (where $r + c \leq 2$):

```
(0,0) (0,1) (0,2)
(1,0) (1,1)
(2,0)
```

`top_k = 5` keeps only the 5 largest. The 6th is **zeroed completely**. This is not quantization noise – it's structured information loss. The dropped coefficient's entire energy becomes reconstruction error.

How much energy that costs depends on the block content. For blocks where all 6 coefficients are significant, the loss is nontrivial.

3. Block Thresholding + Soft Attenuation (`block_threshold`)

The block threshold serves two roles:

Hard gate: Blocks where `max(|AC coefficients|) < block_threshold` are not transmitted. The receiver sees no change – the entire inter-frame difference is lost.

Soft threshold: Blocks that pass the gate have each coefficient attenuated by the threshold amount before quantization:

```
attenuated = sign(x) * max(|x| - block_threshold, 0)
```

This shrinks large coefficients toward zero and zeros out any individual coefficient within a passing block that falls below the threshold. The effect is that `block_threshold` acts as both a noise gate (kill silent blocks) and a noise floor subtraction (remove the noise pedestal from active blocks before quantizing).

- **Threshold too low**: transmit sensor noise as if it were motion. Wastes bytes, adds noise to reconstruction.
- **Threshold too high**: miss real motion. Subtle changes disappear. Attenuation removes too much signal energy from active blocks.

The correct threshold should be set **above the camera's noise floor** so noise blocks are suppressed, but below the weakest real motion signal.

The Measurement Plan

Step 1 – Characterize the Sensor Noise Floor

Capture a sequence of frames with the camera pointed at a **static scene** (no motion). Any inter-frame differences are pure sensor noise.

For each consecutive pair: 1. Compute `diff = frame_B - frame_A` 2. For each 8x8 block, compute `wht_2D(block)` 3. Record `max(|AC coefficients|)` per block

This gives a distribution of noise-only AC magnitudes. The p95 of this distribution is the **noise floor** – the level below which 95% of blocks are pure noise.

`noise_floor_stats()` in `codec.py` already does this. You need static- scene frames to run it on.

Step 2 – Set `block_threshold` from Noise Floor

```
block_threshold = noise_floor_p95 * safety_margin
```

A safety margin of 1.2-1.5x gives some headroom. If p95 = 300, set threshold to 360-450.

If the current `block_threshold = 400` is already above p95, you're fine. If it's below, you're transmitting noise.

Step 3 – Measure Quantization SNR

For frames with **real motion**, run the full encode/decode pipeline and measure:

$$\text{PSNR} = 10 \log_{10}\left(\frac{255^2}{\text{MSE}}\right)$$

Then vary `q_step` and plot PSNR vs. compressed bytes. This traces out the **rate-distortion curve** – the fundamental tradeoff of the codec.

Expected behavior (from the theory): - Halving `q_step`: +6 dB PSNR, ~2x more bytes - Doubling `q_step`: -6 dB PSNR, ~0.5x bytes

Step 4 – Measure Top-K Impact

Fix `q_step`, sweep `top_k` from 1 to 6 (full seq_thresh=2 set). Plot PSNR vs. bytes. This tells you whether the 6th coefficient matters for your content.

Step 5 – Temporal Drift

The decoder accumulates: `frame_B = frame_A + reconstructed_diff`. Each frame's quantization error gets baked into the reference for the next frame. Over N frames, errors can drift.

Measure PSNR of the 1st, 5th, 10th, 20th reconstructed frame relative to ground truth. If PSNR drops significantly over time, temporal drift is a problem and you may need periodic I-frames (send a full reference).

What You Need

1. **Static-scene captures** – 10+ consecutive frames, camera stationary, scene stationary. For noise floor measurement.
2. **Motion captures** – the deer/squirrel sequences you already have. For rate-distortion and top-K sweeps.
3. Run `noise_floor_stats()` on the static frames.
4. Run encode/decode sweeps varying q_step and top_k on the motion frames.

Key Equations Reference

Equation	Meaning
$\sigma^2 = q^2/12$	Quantization noise variance for step size q
$\text{SNR} = 6.02b + 1.76$ dB	Max SNR for b-bit quantizer at full scale
$\text{PSNR} = 10\log_{10}(255^2/\text{MSE})$	Reconstruction quality metric
6 dB per bit	Halving step size = +6 dB = 1 bit of precision

Source: Lyons, *Understanding Digital Signal Processing*, 3rd Ed., Chapter 12, Sections 12.2-12.3.